

A Short Guide to Proposition C.5

The unified Gaussian flat-prior upper bound

Abstract

Proposition C.5 gives one flat-prior upper bound for both Gaussian models. The choice $k = 0$ gives the orbital value $B_{\text{orb}}(\gamma, p)$, while $k = d - 1$ gives the full classical–quantum value $B_{\text{univ}}(\gamma, p) = B_{\text{cl}}(\gamma, d)B_{\text{orb}}(\gamma, p)$. The proof uses the same compact quantum witness in both cases and adds a classical Gaussian witness and scalar convolution argument only when $k = d - 1$. This note states the unified result, gives the detailed full-case argument, and then explains how the orbital case follows by deleting the classical component.

1 The full classical–quantum experiment

The classical space is

$$\mathbf{H} = \left\{ h \in \mathbb{R}^d : \sum_i h_i = 0 \right\}, \quad \dim \mathbf{H} = d - 1,$$

with positive covariance Σ_p . Write

$$f_1 = \varphi_{0, \Sigma_p}, \quad f_\gamma = \varphi_{0, \gamma \Sigma_p}$$

for the centered Gaussian densities. The orbital component lives on the multimode Fock space $\mathcal{F}_{\text{orb}}$ and has centered thermal state τ_p , abbreviated below as τ .

At parameter $\xi = (h, z) \in \mathbf{H} \times \mathbb{C}^{r_d}$, the input and ideal target are

$$\begin{aligned} \Theta_{h,z}^{\text{in}} &= \mathbf{N}_{h, \Sigma_p} \otimes \Phi_z, \\ \Theta_{h,z}^{\text{tar}} &= \mathbf{N}_{\sqrt{\gamma}h, \Sigma_p} \otimes \Psi_z. \end{aligned}$$

The target amplifies both means by $\sqrt{\gamma}$ without increasing either the classical covariance or the centered thermal noise. A physical procedure cannot do this perfectly.

The optimal product procedure dilates the classical observation by $\sqrt{\gamma}$ and applies the product quantum-limited amplifier. Its two losses are

$$B_{\text{cl}}(\gamma, d) = \left(\frac{2\sqrt{\gamma}}{1 + \gamma} \right)^{(d-1)/2}, \tag{1}$$

$$B_{\text{orb}}(\gamma, p) = B(\tau^{(\gamma)}, \tau), \tag{2}$$

and root fidelity multiplies across the classical and quantum factors. The full specialization of Proposition C.5 proves that a joint hybrid channel cannot improve on their product.

2 Statement of Proposition C.5

Proposition C.5 (rephrased). *Let $k \in \{0, d-1\}$. For $k = 0$, take $\mathbf{E} = \{0\}$, omit the classical factor, and interpret integration in h as evaluation at 0. For $k = d-1$, take $\mathbf{E} = \mathbf{H}$ with its induced Euclidean measure. Put*

$$\mathbf{E}_L = \{h \in \mathbf{E} : \|h\|_\infty \leq L\}, \quad \mathbf{Z}_L = \{z : |\Re z_\ell|, |\Im z_\ell| \leq L\}, \quad \Xi_L = \mathbf{E}_L \times \mathbf{Z}_L,$$

with $|\mathbf{E}_L| = 1$ when $k = 0$. Define

$$\begin{aligned} R_{0,z}^{(0)} &= \Phi_z, & S_{0,z}^{(0)} &= \Psi_z, \\ R_{h,z}^{(d-1)} &= \Theta_{h,z}^{\text{in}}, & S_{h,z}^{(d-1)} &= \Theta_{h,z}^{\text{tar}}, \end{aligned}$$

and

$$c_k = \left(\frac{2\sqrt{\gamma}}{1+\gamma} \right)^{k/2}.$$

Then

$$\limsup_{L \rightarrow \infty} \sup_{\Lambda} \frac{1}{|\Xi_L|} \int_{\Xi_L} \mathbf{B}(\Lambda(R_{h,z}^{(k)}), S_{h,z}^{(k)}) \, dh \, dz \leq c_k B_{\text{orb}}(\gamma, p).$$

For $k = d-1$, the supremum includes channels that couple the classical and quantum components.

Since $c_0 = 1$, the right side is $B_{\text{orb}}(\gamma, p)$ in the orbital case. Since $c_{d-1} = B_{\text{cl}}(\gamma, d)$, it is $B_{\text{univ}}(\gamma, p)$ in the full case.

3 The two witnesses

3.1 Classical witness

Define the square-root likelihood ratio

$$\chi(y) = \sqrt{\frac{f_1(y)}{f_\gamma(y)}} = \gamma^{(d-1)/4} \exp\left[-\frac{\gamma-1}{4\gamma} \langle y, \Sigma_p^{-1} y \rangle\right]. \quad (3)$$

It is bounded, strictly positive, and belongs to $C_0(\mathbf{H})$. Moreover,

$$\int_{\mathbf{H}} \frac{f_1(y)}{\chi(y)} \, dy = \int_{\mathbf{H}} \sqrt{f_1(y) f_\gamma(y)} \, dy = B_{\text{cl}}(\gamma, d). \quad (4)$$

The fact that χ vanishes at infinity is as important as the Gaussian identity: it makes integration against χ continuous for weak-* limits of finite measures.

3.2 Quantum witness

In the joint number basis of τ and the quantum-limited output $\tau^{(\gamma)}$, define

$$W_\gamma = \sum_{\mathbf{k}} \sqrt{\frac{t_{\mathbf{k}}}{t_{\mathbf{k}}^{(\gamma)}}} |\mathbf{k}\rangle \langle \mathbf{k}|. \quad (5)$$

It is positive, injective, bounded, compact, and a product of decreasing photon-number weights. Its matching identities are

$$\text{Tr}(\tau W_\gamma^{-1}) = \text{Tr}(\tau^{(\gamma)} W_\gamma) = B_{\text{orb}}(\gamma, p). \quad (6)$$

Lemma C.1 combines these witnesses. If $R(y)$ is a positive classical–quantum density, then

$$B(R, f_1\tau)^2 \leq \left(\int_{\mathbf{H}} \chi(y) \operatorname{Tr}[R(y)W_\gamma] \, dy \right) B_{\text{cl}}(\gamma, d) B_{\text{orb}}(\gamma, p). \quad (7)$$

It remains to show that the first, channel-dependent factor is at most the same product $B_{\text{cl}}B_{\text{orb}}$.

4 Proof roadmap

The proof performs the following reductions:

$$\begin{array}{ccc}
& \text{average fidelity over a large parameter cube} & \\
& \Downarrow & \\
& \text{fidelity at the origin for a Følner-averaged channel} & \\
& \Downarrow & \\
& \text{one weighted moment } M_j & \\
& \Downarrow & \\
\text{quantum compact limit } \Gamma^\chi & & \text{classical scalar limit } \mathcal{R} \\
\text{Lemma C.2} & & \text{Lemma C.4} \\
& \Downarrow & \\
M_j \leq B_{\text{cl}}(\gamma, d) B_{\text{orb}}(\gamma, p) & & \\
& \Downarrow & \\
B \leq B_{\text{cl}}(\gamma, d) B_{\text{orb}}(\gamma, p). & &
\end{array}$$

The two limits come from the same averaged channels but retain different information. The compact quantum limit controls the W_γ moment; the scalar limit controls how much classically weighted trace that quantum map can have. The walkthrough below spells out the case $k = d - 1$. The simplification for $k = 0$ is recorded afterward.

5 Proof walkthrough

Step 1: Følner averaging

For $\xi = (h, z)$, let

$$\begin{aligned}
(\mathcal{U}_\xi^{\text{in}} R)(x) &= D(z)R(x - h)D(z)^*, \\
(\mathcal{U}_\xi^{\text{out}} R)(y) &= D(\sqrt{\gamma}z)R(y - \sqrt{\gamma}h)D(\sqrt{\gamma}z)^*.
\end{aligned}$$

Then

$$\Theta_\xi^{\text{in}} = \mathcal{U}_\xi^{\text{in}}(f_1\tau), \quad \Theta_\xi^{\text{tar}} = \mathcal{U}_\xi^{\text{out}}(f_1\tau).$$

For a hybrid channel Λ , define

$$\bar{\Lambda}_L = \frac{1}{|\Xi_L|} \int_{\Xi_L} \mathcal{U}_{-\xi}^{\text{out}} \circ \Lambda \circ \mathcal{U}_\xi^{\text{in}} \, d\xi. \quad (8)$$

For each input R , this is a Bochner integral in the trace norm of the classical–quantum L^1 space. Strong continuity of translations and Weyl conjugations, together with boundedness of the channels,

makes the integral well defined. For each ξ , the map $\mathcal{U}_{-\xi}^{\text{out}}$ is the inverse output isometry. The definitions of the input and target states therefore give

$$\begin{aligned}
& \frac{1}{|\Xi_L|} \int_{\Xi_L} B(\Lambda(\Theta_{\xi}^{\text{in}}), \Theta_{\xi}^{\text{tar}}) d\xi \\
&= \frac{1}{|\Xi_L|} \int_{\Xi_L} B(\mathcal{U}_{-\xi}^{\text{out}}[\Lambda(\mathcal{U}_{\xi}^{\text{in}}(f_1\tau))], f_1\tau) d\xi \\
&\leq B\left(\frac{1}{|\Xi_L|} \int_{\Xi_L} \mathcal{U}_{-\xi}^{\text{out}}[\Lambda(\mathcal{U}_{\xi}^{\text{in}}(f_1\tau))] d\xi, f_1\tau\right) \\
&= B(\bar{\Lambda}_L(f_1\tau), f_1\tau).
\end{aligned} \tag{9}$$

The first equality uses invariance of root fidelity under the common isometry, the inequality is joint concavity, and the last equality is (8). Joint concavity passes from finite averages to this Bochner integral by trace-norm continuity of root fidelity.

Define

$$F_L(\Lambda) = \frac{1}{|\Xi_L|} \int_{\Xi_L} B(\Lambda(\Theta_{\xi}^{\text{in}}), \Theta_{\xi}^{\text{tar}}) d\xi, \quad F_* = \limsup_{L \rightarrow \infty} \sup_{\Lambda} F_L(\Lambda).$$

Choose $L_j \rightarrow \infty$ so that $\sup_{\Lambda} F_{L_j}(\Lambda) \rightarrow F_*$. For each j , choose a channel $\tilde{\Lambda}_j$ satisfying

$$F_{L_j}(\tilde{\Lambda}_j) \geq \sup_{\Lambda} F_{L_j}(\Lambda) - \frac{1}{j},$$

and define its Følner average by

$$\Lambda_j := \overline{\tilde{\Lambda}_{jL_j}}$$

using (8). To make the translation of the averaging cube explicit, define

$$A_{j,\eta} = \mathcal{U}_{-\eta}^{\text{out}} \circ \tilde{\Lambda}_j \circ \mathcal{U}_{\eta}^{\text{in}}, \quad \Lambda_j = \frac{1}{|\Xi_{L_j}|} \int_{\Xi_{L_j}} A_{j,\eta} d\eta.$$

The group law for the input and output actions and the change of variables $u = \eta + \xi$ give

$$\begin{aligned}
\Lambda_j(\mathcal{U}_{\xi}^{\text{in}} R) &= \mathcal{U}_{\xi}^{\text{out}} \frac{1}{|\Xi_{L_j}|} \int_{\Xi_{L_j} + \xi} A_{j,u}(R) du, \\
\mathcal{U}_{\xi}^{\text{out}} \Lambda_j(R) &= \mathcal{U}_{\xi}^{\text{out}} \frac{1}{|\Xi_{L_j}|} \int_{\Xi_{L_j}} A_{j,u}(R) du.
\end{aligned}$$

When these expressions are subtracted, the integrals over $(\Xi_{L_j} + \xi) \cap \Xi_{L_j}$ cancel. Every $A_{j,u}$ is a channel conjugated by input and output isometries, so $\|A_{j,u}(R)\|_1 \leq \|R\|_1$. The remaining two integrals are supported on the symmetric difference, which gives

$$\|\Lambda_j(\mathcal{U}_{\xi}^{\text{in}} R) - \mathcal{U}_{\xi}^{\text{out}} \Lambda_j(R)\|_1 \leq \frac{|(\Xi_{L_j} + \xi) \triangle \Xi_{L_j}|}{|\Xi_{L_j}|} \|R\|_1 \rightarrow 0. \tag{10}$$

Indeed, $\Xi_L = L\Xi_1$ and Ξ_1 is a bounded polytope, so for fixed ξ the boundary ratio in (10) is $O_{\xi}(L_j^{-1})$.

Step 2: reduce fidelity to one moment

Put

$$R_j = \Lambda_j(f_1 \tau), \quad M_j = \int_{\mathbf{H}} \chi(y) \operatorname{Tr}[R_j(y) W_\gamma] dy.$$

The weighted hybrid inequality (7) gives

$$B(R_j, f_1 \tau)^2 \leq M_j B_{\text{cl}}(\gamma, d) B_{\text{orb}}(\gamma, p). \quad (11)$$

The rest of the proof is devoted to the upper bound

$$\limsup_j M_j \leq B_{\text{cl}}(\gamma, d) B_{\text{orb}}(\gamma, p). \quad (12)$$

Step 3: extract the weighted quantum map

For a trace-class operator X on $\mathcal{F}_{\text{orb}}$, define

$$\Gamma_j^\chi(X) = \int_{\mathbf{H}} \chi(y) [\Lambda_j(f_1 X)](y) dy. \quad (13)$$

This map feeds Λ_j a fixed classical Gaussian with quantum input X , then averages the quantum output with the classical weight χ . It is completely positive and satisfies

$$\operatorname{Tr} \Gamma_j^\chi(X) \leq \|\chi\|_\infty \operatorname{Tr} X \quad (X \geq 0).$$

Notice that $\|\chi\|_\infty = \gamma^{(d-1)/4}$ may exceed one. Hence this initial estimate does *not* make Γ_j^χ trace nonincreasing.

Set $\xi = (0, z)$ and apply (10) to the input $R = f_1 X$. There is then no classical translation, so

$$\mathcal{U}_{(0,z)}^{\text{in}}(f_1 X) = f_1 D(z) X D(z)^*.$$

Multiplying the output density by χ , integrating over its classical variable, and using (13) give

$$\begin{aligned} & \left\| \Gamma_j^\chi(D(z) X D(z)^*) - D(\sqrt{\gamma} z) \Gamma_j^\chi(X) D(\sqrt{\gamma} z)^* \right\|_1 \\ & \leq \|\chi\|_\infty \frac{|\langle \Xi_{L_j} + (0, z) \rangle \Delta \Xi_{L_j}|}{|\Xi_{L_j}|} \|X\|_1 \longrightarrow 0. \end{aligned} \quad (14)$$

Here $\|f_1 X\|_1 = \|X\|_1$ because f_1 is a probability density. The maps Γ_j^χ are uniformly bounded, and (14) is precisely the asymptotic covariance hypothesis of Lemma C.2 with input unitaries $D(z)$ and output unitaries $D(\sqrt{\gamma} z)$. Lemma C.2 therefore gives, after taking a subsequence, a completely positive exactly covariant map Γ^χ with

$$\operatorname{Tr}[\Gamma_j^\chi(X) K] \longrightarrow \operatorname{Tr}[\Gamma^\chi(X) K] \quad (K \text{ compact}). \quad (15)$$

Because W_γ is compact,

$$M_j = \operatorname{Tr}[\Gamma_j^\chi(\tau) W_\gamma] \longrightarrow \operatorname{Tr}[\Gamma^\chi(\tau) W_\gamma]. \quad (16)$$

To apply the least-noise moment to Γ^χ , the proof still has to improve its trace bound from $\|\chi\|_\infty$ to one. This is the purpose of the classical limit.

Step 4: extract the scalar trace kernel

For $u \in L^1(\mathbf{H})$, let $\mathcal{R}_j(u)$ be the classical trace measure of $\Lambda_j(u\tau)$:

$$\mathcal{R}_j(u)(A) = \int_A \text{Tr}[\Lambda_j(u\tau)(y)] \, dy. \quad (17)$$

Since Λ_j is a channel, $\mathcal{R}_j$ is a positive contraction from $L^1(\mathbf{H})$ to the finite Radon measures.

A further diagonal extraction on countable dense subsets of $L^1(\mathbf{H})$ and $C_0(\mathbf{H})$ gives a positive contraction $\mathcal{R}$ such that

$$\mathcal{R}_j(u) \xrightarrow{w^*} \mathcal{R}(u) \quad (u \in L^1(\mathbf{H})).$$

Set $\xi = (h, 0)$ and apply (10) to $R = u\tau$. The input action translates u , while the output action translates the classical output by $\sqrt{\gamma}h$. Hence

$$\begin{aligned} & \|\Lambda_j(u(\cdot - h)\tau) - [\Lambda_j(u\tau)](\cdot - \sqrt{\gamma}h)\|_1 \\ & \leq \frac{|(\Xi_{L_j} + (h, 0)) \triangle \Xi_{L_j}|}{|\Xi_{L_j}|} \|u\|_1 \rightarrow 0. \end{aligned} \quad (18)$$

For $\phi \in C_0(\mathbf{H})$, taking the oscillator trace and pairing the classical output with ϕ therefore give

$$\begin{aligned} & \left| \int_{\mathbf{H}} \phi(y) \mathcal{R}_j(u(\cdot - h))(\,dy) - \int_{\mathbf{H}} \phi(y + \sqrt{\gamma}h) \mathcal{R}_j(u)(\,dy) \right| \\ & \leq \|\phi\|_{\infty} \|\Lambda_j(u(\cdot - h)\tau) - [\Lambda_j(u\tau)](\cdot - \sqrt{\gamma}h)\|_1 \rightarrow 0. \end{aligned}$$

Weak-* convergence applies to the first integral with input $u(\cdot - h)$ and to the second because $y \mapsto \phi(y + \sqrt{\gamma}h)$ belongs to $C_0(\mathbf{H})$. Passing to the limit gives

$$\int_{\mathbf{H}} \phi(y) \mathcal{R}(u(\cdot - h))(\,dy) = \int_{\mathbf{H}} \phi(y + \sqrt{\gamma}h) \mathcal{R}(u)(\,dy).$$

Since $C_0(\mathbf{H})$ separates finite Radon measures, this is the following measure identity. With $(T_a\nu)(A) := \nu(A - a)$,

$$\mathcal{R}(u(\cdot - h)) = T_{\sqrt{\gamma}h} \mathcal{R}(u). \quad (19)$$

Lemma C.4 identifies this covariant scalar kernel: there is a finite positive measure μ , with $\mu(\mathbf{H}) \leq 1$, such that

$$\mathcal{R}(u)(A) = \int_{\mathbf{H}} u(x) \mu(A - \sqrt{\gamma}x) \, dx. \quad (20)$$

Step 5: bound the trace of the weighted quantum limit

Let $P_N \uparrow I$ be finite-rank projections on $\mathcal{F}_{\text{orb}}$. Positivity and (15) give

$$\begin{aligned} \text{Tr } \Gamma^{\chi}(\tau) &= \sup_N \text{Tr}[\Gamma^{\chi}(\tau) P_N] \\ &\leq \liminf_j \text{Tr } \Gamma_j^{\chi}(\tau). \end{aligned} \quad (21)$$

By (13) and (17),

$$\text{Tr } \Gamma_j^{\chi}(\tau) = \int_{\mathbf{H}} \chi(y) \mathcal{R}_j(f_1)(\,dy).$$

Because $\chi \in C_0(\mathbf{H})$, weak-* convergence of the scalar measures passes this integral to the limit. Hence

$$\text{Tr } \Gamma^{\chi}(\tau) \leq \int_{\mathbf{H}} \chi(y) \mathcal{R}(f_1)(\,dy). \quad (22)$$

Use the convolution representation (20). If $X \sim \mathbf{N}_{0, \Sigma_p}$, then $\sqrt{\gamma}X$ has density f_γ , so

$$\int_{\mathbf{H}} \chi(y) \mathcal{R}(f_1)(dy) = \int_{\mathbf{H}} c_\gamma(r) \mu(dr), \quad (23)$$

$$c_\gamma(r) = \int_{\mathbf{H}} \chi(y) f_\gamma(y-r) dy. \quad (24)$$

Completing the Gaussian square gives

$$c_\gamma(r) = B_{\text{cl}}(\gamma, d) \exp \left[-\frac{\gamma-1}{2\gamma(\gamma+1)} \langle r, \Sigma_p^{-1} r \rangle \right] \leq B_{\text{cl}}(\gamma, d). \quad (25)$$

The maximum occurs at zero added classical noise. Since $\mu(\mathbf{H}) \leq 1$, equations (22)–(25) yield

$$\text{Tr } \Gamma^\chi(\tau) \leq B_{\text{cl}}(\gamma, d). \quad (26)$$

This estimate is first proved only at τ . Exact displacement covariance makes it global. Indeed, the positive functional $X \mapsto \text{Tr } \Gamma^\chi(X)$ is represented by a bounded positive operator A . Covariance implies that A commutes with every multimode Weyl displacement, so irreducibility gives $A = cI$. Since $\text{Tr } \tau = 1$,

$$c = \text{Tr } \Gamma^\chi(\tau) \leq B_{\text{cl}}(\gamma, d) \leq 1.$$

Therefore Γ^χ is trace nonincreasing and Proposition C.3 applies.

Step 6: apply the quantum least-noise bound

Using (16), Proposition C.3, and (6),

$$\begin{aligned} \lim_j M_j &= \text{Tr}[\Gamma^\chi(\tau) W_\gamma] \\ &\leq \text{Tr } \Gamma^\chi(\tau) \text{Tr}(\tau^{(\gamma)} W_\gamma) \\ &\leq B_{\text{cl}}(\gamma, d) B_{\text{orb}}(\gamma, p). \end{aligned}$$

This proves (12). Substitution into (11) gives

$$\limsup_j B(R_j, f_1 \tau)^2 \leq (B_{\text{cl}}(\gamma, d) B_{\text{orb}}(\gamma, p))^2.$$

Taking square roots and using the Følner concavity bound (9) proves the full specialization of Proposition C.5.

6 The orbital specialization

For $k = 0$, the Euclidean space is the one-point space $\mathbf{E} = \{0\}$. The conventions in Proposition C.5 read

$$f_1 = f_\gamma = \chi = 1, \quad c_0 = 1, \quad \Xi_L = Z_L.$$

There is no classical integral, trace kernel, or convolution measure. The Følner average is taken only over the displacement variable z , and it is a channel on the oscillator trace class. Lemma C.2 produces an exactly displacement-covariant limit Γ . Its trace-bound clause with $c = 1$ makes this limit trace nonincreasing.

Lemma C.1 and compactness of W_γ give

$$\limsup_j \mathbb{B}(\Gamma_j(\tau), \tau)^2 \leq \text{Tr}[\Gamma(\tau)W_\gamma]B_{\text{orb}}(\gamma, p).$$

Proposition C.3 and $\text{Tr} \Gamma(\tau) \leq 1$ then imply

$$\text{Tr}[\Gamma(\tau)W_\gamma] \leq \text{Tr}(\tau^{(\gamma)}W_\gamma) = B_{\text{orb}}(\gamma, p).$$

Thus the flat-prior fidelity is at most $B_{\text{orb}}(\gamma, p)$. This is exactly the $k = 0$ branch of Proposition C.5. In particular, Lemma C.4 is needed only for the $k = d - 1$ branch, where it supplies the additional factor $B_{\text{cl}}(\gamma, d)$.

7 Why the classical and quantum factors multiply

In the full case, the proof does not assume that the competing channel factors. Instead, factorization emerges at the level of witnesses:

- χ is the square-root likelihood ratio for the two classical Gaussian covariances;
- W_γ is the square-root likelihood-ratio observable for the two thermal laws;
- Lemma C.1 separates the channel-dependent moment from the two fixed inverse-witness integrals;
- Lemma C.4 bounds the trace available to the quantum moment by the classical affinity B_{cl} ;
- Proposition C.3 bounds the normalized quantum moment by the orbital affinity B_{orb} .

Thus any classical–quantum coupling is already included in the single moment M_j , yet the covariance constraints force that moment to pay both losses.

8 Common points of confusion

1. The compact-limit lemma alone gives $\text{Tr} \Gamma^\chi(X) \leq \|\chi\|_\infty \text{Tr} X$, which is not enough for Proposition C.3. The scalar convolution calculation is what improves the coefficient to $B_{\text{cl}} \leq 1$.
2. Two different compactness arguments are used. Lemma C.2 treats quantum maps through compact operators, while the diagonal extraction of $\mathcal{R}_j$ treats classical finite measures through $C_0(\mathbf{H})$.
3. Compactness of W_γ passes the quantum moment to the limit; membership $\chi \in C_0(\mathbf{H})$ passes the classical weighted trace to the limit. These are parallel requirements.
4. The noise measure μ need not be normalized or absolutely continuous. The estimate uses only positivity and $\mu(\mathbf{H}) \leq 1$.

9 Role in the paper

The two specializations of Proposition C.5 provide the upper bounds for both parts of the Gaussian amplification theorem. Quantum-limited amplification attains $B_{\text{orb}}(\gamma, p)$ in the orbital model. Classical dilation combined with quantum-limited amplification attains

$$B_{\text{univ}}(\gamma, p) = B_{\text{cl}}(\gamma, d)B_{\text{orb}}(\gamma, p)$$

in the full model. LAN fidelity transfer carries the orbital converse to known-spectrum cloning and the full converse to unknown-spectrum cloning, where the additional classical penalty appears.